

Tayden White

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EDUCATION

Purdue University | West Lafayette, IN

Aug 2023 - May 2027

Dual Degrees in Computer Science & Mathematical Economics

GPA: 3.85 – Dean's List & Semester Honors (6x)

Relevant Coursework: Data Structures & Algorithms, Data Mining & Machine Learning, Game Theory, Econometrics

Undergraduate Research | West Lafayette, IN

Communication and Cognition Lab @ Purdue University

Mar 2026 – Current

- Built a full-stack web application in Python using Streamlit to facilitate an experiment in which one human and 2-3 LLMs deliberate over three Hidden Profile Paradigm group decision-making tasks
- Automated transcript coding with an LLM extraction pipeline, using the OpenAI API to classify information-sharing events across ~2000 HPP discussions, increasing hand-coding comparative accuracy from 60% to 99%
- Contributions appearing in **three forthcoming papers and a book chapter**

Department of Mathematics @ Purdue University

May 2025 - Aug 2025 | May 2026 – Current

- Diagnosed and resolved edge cases in a fusion algorithm for “permutation defects”—quasiparticles which can encode qubits in **topological quantum computing**—producing a revised set of rules to be used in a **future publication**
- Contributed core fusion-ring operations (Deligne product, defect fusion) to an open-source Julia package implementing Fusion Rings

Johnson & Johnson – Innovative Medicine | Raritan, NJ

Software Engineer Co-op

Aug 2025 - Dec 2025

- Optimized an FDA compliance analytics pipeline to support more than 60k records in static Hugo dashboards, cutting build time from ~30 minutes to ~5 minutes
- Designed an agentic text-to-SQL workflow that decomposes natural language questions into parallel queries across Delta tables supporting Hugo site backends

Data Science Intern

May 2024 – Aug 2024

- Refactored an internal “generic” Machine Learning R package into modular components, eliminating 200+ lines of duplicated logic and enabling downstream teams to extend it without forking
- Spearheaded an initiative for interns to build and deploy static sites using Hugo & GitHub Pages, producing tutorial documentation to teach 20+ coworkers how to deploy their own site

PROJECTS

The Future of AI at Purdue | *Python, pandas, matplotlib*

- Surveyed Purdue administrators on classroom AI policy, benchmarking sentiment against peer institutions to quantify divergence; authored a paper visualizing results and recommending Purdue's path forward
- Presented findings to 60+ faculty and students at **Purdue's Undergraduate Research Exposition**

Multithreaded Client-Server Marketplace App | *Java, Swing, TCP Sockets*

- Built a multithreaded Java client-server marketplace over a custom TCP protocol on raw sockets (thread-per-client for concurrent sessions), with a 1,000-line Swing GUI for auth, seller storefronts, search, cart, and order history
- Co-developed in a 5-engineer team across 150+ Git commits, writing and executing a 15+ scenario end-to-end test plan

Unix Shell | *C/C++, Lex, Yacc, Linux*

- Built a Unix shell parsing commands into an executable AST via Lex/Yacc, supporting fork/exec process management, pipes and file-descriptor redirection, signal handling, subshells, and wildcard expansion

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL (PostgreSQL), R, Julia, x86-64 Assembly, Bash

ML & Data: PyTorch, scikit-learn, pandas, NumPy, PySpark, matplotlib, OpenAI API, Databricks/Delta Lake

Frameworks & Tools: Streamlit, Django, React, WebSockets, Hugo, Git, GitHub Actions, Docker, AWS, Linux, LaTeX